

the state-of-the-art research and development in the construction sector, unlike many developed countries and even some of the developing countries which have sophisticated modern construction of building stock. Special efforts will be made in the areas of institutional strengthening and R & D in the construction sector to improve the earthquake-resistant features as this is required due to the booming construction requirements of a fast growing and increasingly urbanized economy.



Traditional Building Typologies

It has been seen in most of the earthquake-prone areas that traditional housing construction in such areas is often influenced by the earthquake risk and traditional artisans have built houses which can withstand the after effects of earthquakes. In several villages of Gujarat, traditional houses called Bhongas withstood the devastating impact of the Bhuj earthquake. Studies are proposed to be carried out in appropriate building typologies which demonstrate the traditional wisdom of construction in earthquake-prone areas, like the Dhajji-Diwari system in Jammu and Kashmir, Bhongas in the Kutch region of Gujarat, brick-nogged wood frame construction in Himachal Pradesh and ekra constructions in Assam.

National School Safety Project

In collaboration with the Ministry of Human Resource Development, NDMA is initiating a Pilot Project for National School Safety in selected high and very high risk districts in the country falling in seismic zones IV and V. Structural Safety Audit of selected school buildings in the selected districts will be carried out for assessing the ability of these structures to withstand high intensity earthquakes and a most vulnerable school building will be selected with the help of the district administration to carry out the retrofitting and seismic strengthening of that structure. It is also envisaged that a National School Safety Policy will be prepared and a Handbook on School Safety will be developed with checklists and formats for conducting structural safety audits and retrofitting of school buildings. Simultaneously, special efforts will be made to improve the capacity building of engineers, education officials, school

children, teachers and parents and for creating greater public awareness on school safety.

Medical Preparedness and Mass Casualty Management

There is an urgent need to strengthen the medical preparedness and mass casualty management facilities in earthquake-prone areas. With the help of the Ministry of Health and Family Welfare, Government of India and the concerned state governments in earthquake-prone states, it is proposed to ensure the availability of Heli-Ambulances, Containerised Mobile Field Hospitals, Improved Bio Safety Laboratories, Integrated Ambulance Network, Trauma care Network, etc. with special emphasis on very high risk earthquake-prone areas. It is expected that many of these activities will be initiated during the Eleventh Five Year period and further intensified subsequently as mandated in the National Disaster Management Guidelines for Medical Preparedness and Mass Casualty Management prepared by NDMA.

Improved Emergency Response Capacity

Selected personnel from Eight battalions of the National Disaster Response Force (NDRF) based in different strategic locations of the country have been trained in Collapsed Structure Search and Rescue, watersmanship training, heli-slithering, Medical First Response, etc. and are being equipped with the state-of-the-art life saving equipments. Various stakeholder groups like elected representatives of the Panchayati Raj Institutions (PRIs), Self Help Groups, National Social Service, Nehru Yuvak Kendra Sangathan (NYKS), Indian Red Cross Society, Non-Governmental Organisations, corporate sector, media and community at large are being trained on various aspects of Community Based Disaster Management. Civil Defence is being revamped to play a significant role in disaster management. Fire and Emergency Services is being strengthened with the provision of modern life saving equipments to play a more active role in multi-hazard response.

All the above activities will improve the earthquake preparedness in India and create an enabling environment for a culture of preparedness, mitigation and improved emergency response capability in the country, to strengthen our resolve to work towards a disaster-resilient India.



Prof. N. Vinod Chandra Menon



The author is Member of
National Disaster Management Authority
Government of India

